A cherry tree ripe for picking: The relationship between the acute:chronic workload ratio and health problems

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This study protocol was approved by the South-Eastern Norway Regional Committee for Medical and Health Research Ethics (2017/1015) and the Norwegian Centre for Research Data (5487). We declare no conflict of interest or financial involvement in any organization or entity with financial interest in the subject of matter discussed in the article.

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ABSTRACT

Objective: To investigate whether the relationship between the acute:chronic workload ratio (ACWR) and health problems varies when different methodological approaches are used to quantify the relationship.

Design: Prospective cohort study

Methods: An online questionnaire was used to collect daily health and training information from 86 elite youth footballers for 105 days. The relationship between players' training load and health was analysed using a range of different definitions of ACWR and health problems. We used a 21-day and 28-day chronic period, coupled and uncoupled calculations, and exponentially weighted moving average (EWMA) and rolling average (RA). ACWR data were categorized into three bins (low, medium and high) using pre-defined categories and z-scores, and we compared medium to high, medium to low and low to high categories. The outcome was defined in three ways: "all health problems", "all injuries", and "new non-contact injuries". We performed random-effects logistic regression analyses of all combinations, a total of 108 analyses.

Results We recorded 6250 training days and 196 health problems. Of the 108 analyses performed, 24 (22%) identified a statistically significant ($p<0.05$) association between ACWR and health problems. A greater proportion of significant associations were identified when using an EWMA (42% of analyses), when comparing low and high categories (33%), and using "all health problems" definition (33%).

Conclusions: The relationship between ACWR and health problems was dependent on methodological approach.

Keywords: *ACWR, Football, Injury, Training Load*

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INTRODUCTION

The acute:chronic workload ratio (ACWR) was introduced by Hulin and co-workers in 2014[31] as a modification of Banister's fitness-fatigue model [5]. It is calculated by dividing the total sum of all training an athlete has recently completed (normally the previous week) by the sum they have completed over a longer period (e.g., the previous 4 weeks). ACWR is intended to reflect athletes' preparedness for training by accounting for both positive and negative training effects (i.e. fitness and fatigue) [25]. The relationship between ACWR and health problems has emerged as a popular topic among practitioners and researchers alike; since 2014, at least 45 cohort studies and multiple reviews have addressed the topic (supplementary #1). The quality of the evidence supporting ACWR has been questioned [36], and several methodological aspects of ACWR calculation and analysis have been debated [11, 20, 30, 43, 44, 52, 60, 66, 71, 73].

There is no consensus on how the relationship between ACWR and health problems should be calculated and analysed, and no two studies have used the same approach. Although most studies report significant findings (supplementary #1), their results and conclusions are inconsistent, and few studies consider their negative findings when making conclusions. The inconsistent results among ACWR studies may reflect differences in methodology, particularly (1) how training load and health problems were recorded, (2) how ACWR was calculated, and (3) how the relationship between ACWR and health problems was analysed.

We aimed to analyse the relationship between ACWR and health problems among elite youth football players using a wide range of methodological combinations defined before analysis. We considered the extent to which these

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methodological choices influenced the relationship between ACWR and health problems.

METHODS

We conducted a prospective cohort study from July to October 2017. During this period, which included one-two weeks of summer break and 13-14 weeks of regular in-season competition, players recorded their daily training load and health status using an online survey software (Briteback AB, version 2.5.3.1; Norrköping, Sweden). We obtained approval from the South-Eastern Norway Regional Committee for Medical and Health Research Ethics (2017/1015) and the Norwegian Centre for Research Data (5487) before study start. All participants or their guardians provided written informed consent. Participants were not involved in the design, conduct, interpretation, and/or translation of this paper.

Recruitment

We invited a convenience sample of 12 (six girls' and six boys' teams) Norwegian football (soccer) teams from the elite U-19 divisions to participate. Teams that already used a training load management system were excluded (n=6), as their data could potentially be non-comparable to previous studies. Six teams (3 girls' and 3 boys') accepted the invitation to participate in the study, and all players in these teams agreed to participate. We invited all permanent squad members of these teams, except players with existing health problems that we expected would render them unavailable for football training and match play during the study period (as they would not provide any data). No power or sample size calculation was performed before the study.

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Collection of training load data

Players reported their daily training load using a short online questionnaire (supplementary file #2). A link to the questionnaire was distributed by an automated short message service (SMS) at 9 p.m. every evening. If players had not replied to the questionnaire before 8 a.m. the following morning, they received an SMS reminder. The questionnaire included questions with structured response options on training load data for all football activity, including organized training and matches, as well as non-organized football play (supplementary #2). Players were asked to record the duration in minutes and their overall perceived rate of physical exertion (RPE) using the modified Borg CR-10 scale [23]. We calculated an arbitrary training load unit (AU) by multiplying the duration and the session RPE (sRPE) [23] for all football activity.

Collection of health data

Players recorded their health status in the daily questionnaire by answering two questions, (1) have you experienced a new health problem and, (2) have you experienced a worsening of an existing health problem? Players were asked to report all complaints, irrespective of their consequences on football participation or their need to seek medical attention [24]. If the player reported any new health problems or a worsening of an existing problem, a sports medicine clinician (physiotherapist or chiropractor) contacted them by telephone the following day to conduct a structured interview. During this interview, we classified health problems according to the Union of European Football Associations (UEFA) guidelines as injury or illness, acute or overuse, contact or non-contact, injury mechanism, tissue and body part (supplementary #3) [34].

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Efforts to limit potential sources of bias

We standardized sRPE-registration by collecting data at 9 p.m., when football activity was likely to be finished, and when the player was unlikely to be in the presence of their teammates. To maximize the likelihood of honest reports, we assured players that all their training and health data would be treated as confidential and not disclosed to their coach or teammates. Players were familiarised with the RPE scale during information meetings before the study, and we instructed players to consider the whole session when giving a RPE score. We contacted coaches weekly to engage their support in motivating their players to respond to the questionnaire. Any player who consistently failed to respond to the automated reminder-SMS was sent personal messages to encourage them to resume participation. To ensure consistent reporting of all health problems, we familiarized players with the definitions in the pre-study meeting and emphasized the importance of reporting all health problems, irrespective of their consequences.

The questionnaire has not been validated. At the time of data collection, there were no validated alternatives. However, 10 players from the same population pilot tested the questionnaire for three weeks and established a high face validity.

To assess the potential bias in our results, we conducted an anonymous post-study survey of all participants, asking whether the players had reported all training and health data as instructed.

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Variables

All independent (training load) and dependent (health status) variables were defined a priori.

Independent variables

ACWR was calculated in eight different ways, using all combinations of the following:

- The “acute” and “chronic” workload periods were defined in two ways: 7 to 28 days and 7 to 21 days.
- Training load was calculated for each period using a simple rolling average (RA) and exponentially weighted moving average (EWMA; for formulas, see supplementary file #4) [71].
- For RA calculations, we included the acute period in the chronic period (coupled approach) and separated it from the chronic period (uncoupled approach). As the original EWMA formula is coupled by nature[71] and to limit the number of analyses, no uncoupled approach was used.

ACWR variables were then converted into ordinal categorical variables with two methods:

- Pre-defined bins using the most common categories: Low (<0.8), medium (0.8 to 1.5) and high (>1.5)[31, 39].
- Z-scores: Low (<1 standard deviation (SD) below the mean), medium (1 SD below the mean to 1 SD above the mean), high (>1 SD above the mean).
    - Z-score categories for RA was: Low (<0.4), medium (0.4 to 1.81) and high (>1.81)
    - Z-score categories for EWMA was: Low (<0.32), medium (0.32 to 1.73) and high (>1.73)

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Dependent variables

Using players' daily health reports, we created three different dependent variables for analysis:

- *Any worsening in health status (yes/no):* all new injuries and illnesses, and all cases when players reported a worsening of an existing injury or illness
- *Any worsening in injury status (yes/no):* all new injuries and cases when players reported a worsening of an existing injury. Illnesses were excluded.
- *Any new non-contact injury (yes/no)*

Statistical analyses

All statistical methods were defined before analysis. A random-effects logistic regression model was used to investigate all combinations of the 12 independent and 3 dependent variables, comparing the medium ACWR group to the low and high groups, and the low group to the high group. We performed a total of 108 (all combinations X comparisons) separate analyses using Stata software (version 15.3 StataCorp LLC, College Station, TX), with the xtlogit command (see supplementary file #5 for script). A random-intercept model was used, and the random error term was assumed normal distributed (with mean 0). We did not adjust the regression models for confounding by sex, age or for the effect of clustering by individuals and teams. However, the adjusted analyses are available as supplementary material (#5)**.** The incidence rate was calculated by dividing the total number of cases satisfying each health problem definition by the total exposure time and multiplying by 1000 hours. We calculated the absolute daily risk of health problems by dividing the number of cases satisfying each health problem definition by the total number of athlete days in the study.

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Missing data

When athletes' training load data were incomplete, either due to missing session duration or RPE, we replaced the missing values of individual players with the team average for that session. If an entire session was unreported, we did not attempt to estimate missing data. Instead, we defined a minimum amount of information necessary to make an ACWR calculation (5 days for a 7-day acute period, 14 days for a 21-day chronic period and 21 days for a 28-day chronic period) and used a statistical model (random-effects logistic regression) capable of handling incomplete datasets.

RESULTS

A total of 86 players agreed to participate, 47 males (mean age 17.5, SD 0.7), and 39 females (mean age 17.5, SD 1.3). The overall response to the daily questionnaire was 69% (6250 of 9030 athlete days).

Training load and exposure

The median weekly training load per player was 1262 AU (interquartile range; IQR [1024, 1422]) for training and 439 AU [425, 536] for matches. The total exposure during the study, including both training and matches, was 4668 hours. The median weekly exposure was 242 [155, 329] training minutes and 64 [47, 81] match minutes, a total of 306 minutes [274, 338] per player.

Health problems

Insert table 1 about here

A total of 196 health problems were reported. Of these, 91 were classified as injuries, 17 as illness, and 88 were not classified. During the study period, 65 players

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(75% of all players) reported a health problem, 62 players (72%) registered an injury, and 39 (45%) reported a new non-contact injury. The incidence rate and daily risk of health problems are shown in Table 1.

Insert table 2 about here

Main results: Association between ACWR and health problems

Of the 108 analyses performed, we found 24 (22%) significant associations between ACWR and health problems, spread across various methodological combinations (figure 1). We did not observe any patterns of combinations that substantially increased the chance of a significant association. The single most frequent component among the significant combinations was EWMA with 15 of the 24 significant associations. The strongest association observed was with the following combination: a coupled 7-day acute, 21-day chronic time-period using EWMA, and by pre-defined categories comparing low to high category with all injuries as health problem definition (OR: 5.27 95% CI: 1.83-15.24).

Insert figure 1 about here

Post-study survey

Forty-five players (53%) responded to the post-study survey. Of these, 15 (33%) players said they had not reported training load data every day as instructed. Participants described the two main reasons why they did not report training load data: (1) they had forgotten to reply to the SMS (9 players), and (2) that it was too time-consuming (two players). Thirteen players (29%) admitted they had not reported all health problems, with the most frequent explanations that they did not think the

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problem was significant enough (eight players) and that they did not want to be contacted by phone for the follow-up interview (three players).

DISCUSSION

We aimed to explore the effect on different methodological choices on the association between ACWR and health problems. As this was a critique of current methods in ACWR research, we based our choice of methods on the published literature and do not necessarily endorse their future use. We analysed 108 different methodological combinations and found 24 (22%) significant associations between ACWR and health problems. There was a large variation in the size and precision of the estimated strength of the association. Our main conclusion is that the potential association between ACWR and health problems is affected by methodological choices.

Health problem definition

The definition of a health problem had the largest effect on the number of significant associations identified between ACWR and health problems. When analysed with the broadest definition, almost one in three analyses was significant, but when using the narrower definitions all injuries and new non-contact injuries, the proportion of significant findings was reduced (25% and 2% respectively). One likely explanation is that the number of cases was reduced from 196 to 45 when we applied the most narrow health problem definition. This is likely to be too few cases to detect a small to moderate association (which requires approximately 200 cases) [4]. As only six out of the 43 studies that provided number of cases (supplementary #1)

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included more than 200, a large proportion of studies in this field appear to be underpowered.

Most ACWR studies have used a time-loss health problems definition [7-9, 15, 17-19, 21, 22, 29, 31, 32, 38, 40, 41, 45-48, 50, 54, 55, 57, 58, 61-64, 67, 68, 74]. However, the narrow definition is unlikely to capture a majority of overuse injuries, which do not often lead to time loss [3, 13, 14]. As overuse injuries are likely the injury type most associated with training load, we believe it is essential to use a broad health problems definition in future training load studies.[12, 13]

ACWR calculation

The duration of the acute and chronic time-periods varies considerably between the previous ACWR studies. Time-periods are often sport-specific and typically aim to represent one training micro-cycle (acute) and one macro-cycle (chronic)[10]. We chose the two most used chronic periods, 28 days and 21 days and found that 28 days had 50% more significant associations than 21. Most studies define the acute period as 7 days (supplementary #1); however, some define it differently (e.g. 3 day)[1, 10]. We did not investigate different acute period durations to limit the number of analyses.

Typically the average acute and chronic workloads are determined using a simple rolling average of the entire period [7-9, 16-19, 21, 22, 29, 31-33, 37, 39-42, 45-47, 51, 53, 55-59, 61, 63-65, 67-69, 74]. However, Williams et al.[71] proposed the use of an EWMA, whereby each day's contribution to the aggregate is exponentially greater than the previous day. Several studies have shown EWMA-based ACWR calculations to be more closely associated with health problems than RA-based calculations [21, 27, 39, 55, 62]. Our findings support this, with EWMA being significant in 42% of the analyses.

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Recently, Lolli et al.[43, 44] criticized the use of coupled ACWR, as it by definition leads to a correlation between the numerator and the denominator in the ratio. We found almost 50% more significant associations when using a coupled ACWR than non-coupled, suggesting that the two cannot be directly compared.

ACWR analysis

Almost all training load data are converted from a continuous to a categorical variable for analysis. Recently, Carey et al. highlighted problems with large false discovery and rejection rates with this approach [11]. Loss of power and residual confounding are other important problems with discretization of continuous variables [2, 6], and cannot be recommended in ACWR research. Despite the challenges, few attempts have been made to investigate ACWR as a continuous variable [41] . Instead, studies have categorised ACWR values according to the first ACWR publication [31], or by z scores. We categorised data in two ways, using pre-defined cut-off values and using a z-score, and there was not a large difference between the two. Rather, the limitations of each approach should be recognised. In small sample sizes, or in any sample size that discretises into small brackets, sparse data is likely to affect results,[26, 66] and especially in the low-categories where data are likely to be limited. This is apparent in our results when considering the inconsistency in the OR estimates particularly in the comparisons that included the low category. The analysis of categories with low power provides uncertainty and should be interpreted with caution.

Another consideration with discretizing is the number of categories and the comparison between these. The number of categories and comparisons vary across the literature (supplementary #1). We chose to compare a medium category to high

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and low, and the low category to high. In both cases involving an increased ACWR we found significantly higher odds of developing a health problem in almost 1 in 3 analyses. In contrast, we only identified two significant associations when comparing medium to low ACWR. This finding contrasts with the originally-described ACWR concept, which suggested an increased injury risk when ACWR was low [25]. We note that few (n=2) [63, 69] other studies have supported this aspect of the concept, and question its biological rationale.

We believe that discretisation is inappropriate in ACWR-injury studies and a method for modeling training load as a continuous variable is warranted.

Statistical approach

A variety of statistical models have been used to analyse the workload between ACWR and health problems. Most studies have used an overly-simple approach, without accounting for the multifactorial aetiology of injuries, between-athlete or within-athlete assessments, missing and unbalanced data, and the nature of time-varying variables [72]. To tackle some of these challenges, we chose a random-effects logistic regression model, which is one of two recommended approaches [72]. Nevertheless, we believe that research comparing the results of different statistical approaches is warranted.

Methodological considerations

We collected health data directly from athletes; a design that has not previously been used in ACWR studies. This approach allowed us to use a broad definition of health problems, which means we are likely to have captured a far greater proportion of overuse injuries than previous studies [12, 13]. However, this

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approach also has limitations. Health problems were not medically confirmed, lacked diagnostic details, and in many cases were not fully classified (e.g. as acute or overuse injuries). Furthermore, players failed to respond to 31% of the daily questionnaires and, as illustrated by our post-study survey, did not always report every health problem or training session, potentially leading to selection bias.

Session-RPE is considered a validated method for measuring training load across various sports [28], and for elite youth footballers specifically [35]. We chose sRPE because it was the most practical way to quantify load in footballers of this level and because a large majority of previous ACWR studies have used it as their primary load measure.

The handling of missing data might affect our results, and we used a sub-optimal method where the team average replaced missing values when either duration or RPE were missing. Although this is the most common method used in previous studies, future research should investigate better options, such as the methods discussed in White et al[70]. We cannot exclude the possibility that the discretization of ACWR has influenced results through a loss of data, false discovery and rejection rates, residual confounding and sparse data bias [26].

Future directions

All previous ACWR studies have used different methodological approaches and their findings are inconsistent. In general, the conclusions of these studies have focussed on significant findings. We have demonstrated the number of non-significant findings likely to be behind every positive one. Reporting bias is therefore an important consideration when assessing the overall evidence for the relationship between

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ACWR and health problems. Although each sport and study population are different, methodological consensus for workload studies is needed.

Finally, we question the need for more underpowered risk factor studies examining the relationship between training load and health problems. We believe that to improve our understanding of the relationship between ACWR and health problems; it is necessary to conduct much larger observational studies (i.e. thousands and not hundreds of injuries), as well as experimental research such as randomised controlled trials.

Conclusion

The potential association between ACWR and health problems is affected by methodological choices. Future ACWR studies should pre-register their definitions, hypotheses and models, and report the results of all performed analyses.

KEY POINTS

Findings

The relationship between ACWR and health problems depends on methodological choices.

If investigators cherry-pick positive results (i.e. only report significant findings, rather than the results of all analyses) an overestimation of the association between ACWR and health problems is expected.

Implications

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In future studies of training load and health problems, we encourage researchers to report all analyses performed, as well as their rationale for all methodological choices.

Different study designs are needed to test whether using ACWR to manage athletes' training load is an effective injury prevention measure.

Practitioners should be cautious when comparing studies using different methodological approach and be aware the possibility that authors may choose to report only the results that support their own hypotheses.

Caution

A limitation of this study is the use of sub-optimal methods from previously published articles. This was done to compare the difference of previous approaches. We did not intend the results to be interpreted to evidence for the relationship between ACWR and health problem itself.

Declarations

Acknowledgments

The Oslo Sports Trauma Research Center has been established at the Norwegian School of Sport Sciences through generous grants from the Royal Norwegian Ministry of Culture, the South-Eastern Norway Regional Health Authority, the International Olympic Committee, the Norwegian Olympic Committee and Confederation of Sport, and Norsk Tipping AS. The authors are grateful to the assessors, and to all the players and coaches who participated in the study.

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The authors would like to acknowledge Adam Sullivan in helping us with initial EWMA calculations and Eirik Halvorsen Wik for valuable feedback on the draft manuscript.

Funding

The Oslo Sports Trauma Research Center has been established at the Norwegian School of Sport Sciences through generous grants from the Royal Norwegian Ministry of Culture, the South-Eastern Norway Regional Health Authority, the International Olympic Committee, the Norwegian Olympic Committee and Confederation of Sport, and Norsk Tipping AS

Conflict of interest /Competing Interests

None

Availability of data and material

All raw data are available in supplementary files

Authors contribution

TD, BC, TEA, JB, and MWF planned the study. The data collection was done by TD, KNM, MV, and MK. All authors have been involved in the data analyses, drafting, and revision of the manuscript, and all have approved the final version.

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TABLE 1. *Health problems (All health problems, All injuries, New non-contact injuries), incidence rate, and absolute daily risk*

Health problem definition	N	Incidence per 1000h	95% CI	Absolute daily risk	95% CI
All Health problems	196	42.0	36.3 - 48.3	3.14 %	2.72 – 3.6
All Injuries	91	19.4	15.6 - 23.9	1.46 %	1.17 – 1.78
New non-contact injuries	45	**9.63**	7.9 – 12.9	0.72 %	0.53 – 0.96

TABLE 2*: Number of* ***significant*** *associations and number of analyses based on variables. Abbreviations: EWMA, exponentially weighted moving average; NNC-Injuries, new non-contact injuries.*

	N of analyses	N of significant associations	OR Range med to low	OR Range med to high	OR Range low to high
Average					
Rolling average	72	7 (10%)	0.24 to 7.92	0.42 to 5.19	0.16 to 6.78
EWMA	36	15 (42%)	0.16 to 1.58	0.57 to 7.44	0.63 to 15.25
Chronic time-period					
28-days	54	8 (15%)	0.02 to 7.92	0.51 to 5.19	0.17 to 12.9
21-days	54	14 (26%)	0.15 to 5.82	0.42 to 7.44	0.17 to 15.25
Coupling					
Coupled	72	15 (21%)	0.15 to 7.92	0.42 to 7.44	0.17 to 15.25
Non-coupled	36	7 (19%)	0.27 to 7.29	0.49 to 5.19	0.18 to 6.85
Categories					
Pre-defined	54	12 (22%)	0.22 to 2.46	0.42 to 7.44	0.37 to 15.25

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Z-Score	54	10 (19%)	0.15 to 7.92	0.47 to 5.19	0.17 to 13.32
Comparison					
Med to low	36	2 (6%)	n/a	n/a	n/a
Med to high	36	8 (22%)	n/a	n/a	n/a
Low to high	36	12 (33%)	n/a	n/a	n/a
Health problem definition					
All health problems	36	11 (31%)	0.24 to 2.05	0.73 to 3.34	0.75 to 6.85
All injuries	36	9 (25%)	0.17 to 6.29	0.58 to 5.64	0.27 to 15.25
NNC-Injuries	36	2 (6%)	0.15 to 7.92	0.42 to 7.44	0.17 to 14.45

FIGURE 1. *Analyses of the relationship between ACWR and all health problems, all injuries and new non-contact injuries. For all details, see supplementary material #5. Abbreviations: C, coupled; NC, non-coupled; RA, rolling average; EWMA, exponentially weighted moving average; Z, z-score based categories; PD, pre-defined categories*

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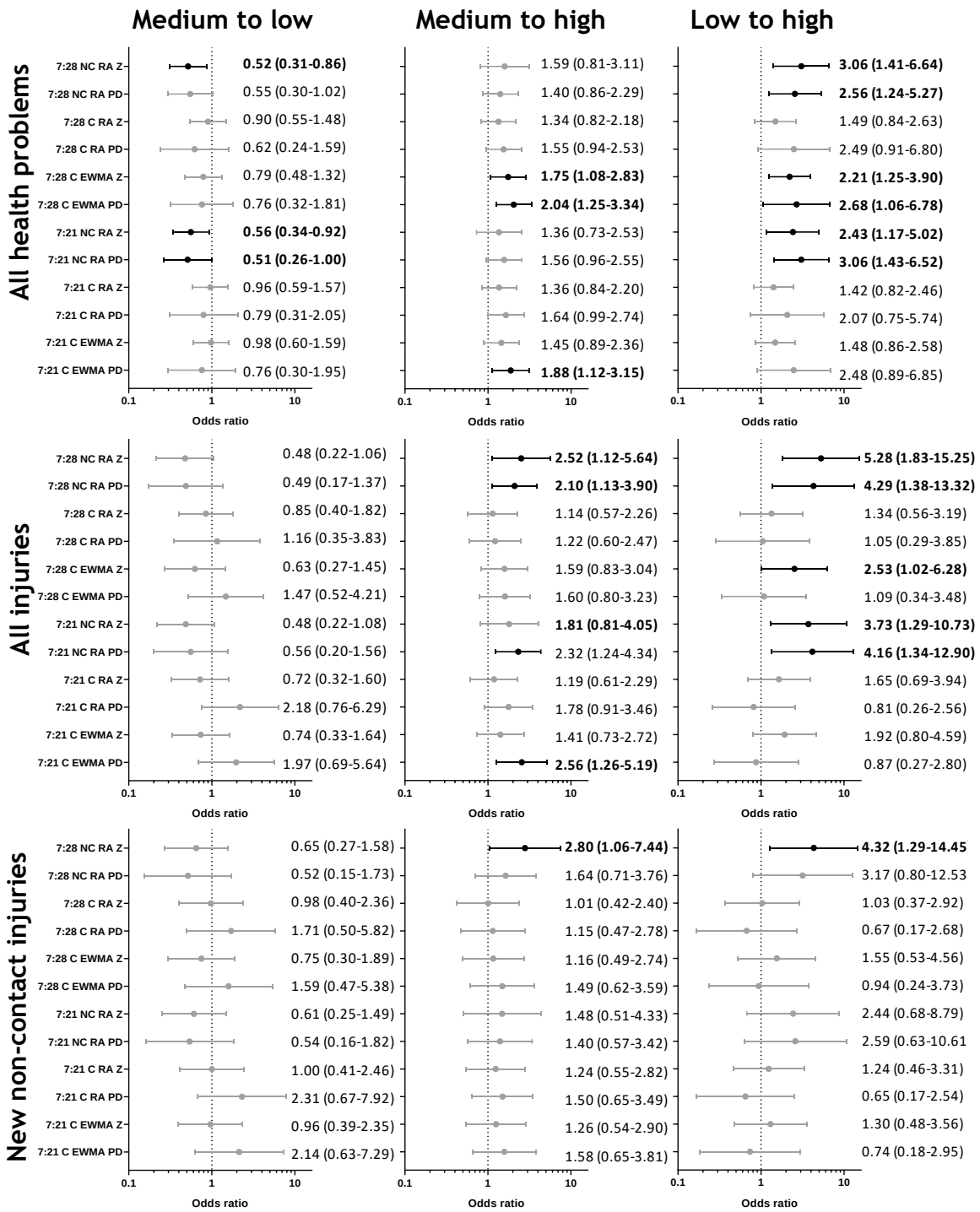

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